

Gate-All-Around (GAA) nanosheet FETs (December 2025)

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Gate-All-Around (GAA) nanosheet field-effect transistors have emerged as a leading solution for continued CMOS scaling beyond the limits of FinFET technology. Due to their superior electrostatic control, vertically scalable channel architecture, and strong performance-power tradeoffs, they are well-suited for advanced logic and memory applications at sub-5 nm technology nodes. This paper examines recent advances in GAA nanosheet transistor technology and how these devices compare to alternative architectures such as TreeFETs and CombFETs. The fundamentals of the device are reviewed, focusing on the stacked nanosheet structure and full gate enclosure. Recent studies demonstrate that nanosheet FETs achieve significantly higher drive current and improved energy efficiency through vertical channel stacking and optimized sheet width [1]-[4]. Circuit-level performance advantages are also explored, showing improved switching speed and reduced energy-delay product relative to competing architectures. Future scaling pathways are discussed through a deep dive into how drive current scales with nanosheet width and stacked-channel count, highlighting why nanosheet FETs outperform TreeFETs and CombFETs in modern integrated circuits.

I. DEVICE FUNDAMENTALS

THE Gate-All-Around (GAA) nanosheet field-effect transistor developed to extend transistor scaling beyond the physical and electrostatic limits of FinFET technology. Unlike standard planar MOSFETs and FinFETs where the gate only partially controls the channel, the GAA architecture fully surrounds the channel with the gate. This complete enclosure provides superior electrostatics, suppresses short-channel effects, and allows continued scaling at sub-5 nm technology nodes [1], [3], [8]. Due to this enhanced control and its vertically stacked channel structure, the GAA nanosheet FET has become the leading transistor architecture for high performance and low power integrated circuits.

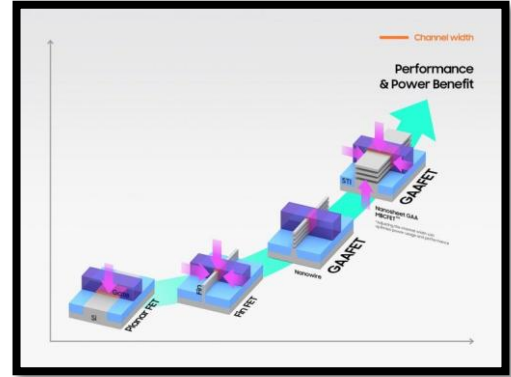


Figure 1: Technology evolution from planar MOSFETs to FinFETs and gate-all-around (GAA) nanosheet FETs, illustrating the increase in effective channel width and corresponding improvements in performance and power efficiency at advanced technology nodes [6].

II. STRUCTURE AND FUNCTION

The structure of a Gate-All-Around (GAA) nanosheet field-effect transistor is designed to maximize electrostatic control while enabling continued device scaling. Unlike FinFETs, which use a single vertical fin as the conduction channel, the GAA nanosheet FET consists of multiple horizontally stacked nanosheets that serve as parallel current conducting channels. Each nanosheet is completely surrounded by the gate electrode, separated by a thin gate dielectric. This full gate enclosure allows for uniform control of the channel from all directions and significantly reduces short-channel effects.

The typical device structure includes a source and drain region that contact the ends of the nanosheets, a stacked channel region suspended between them, a dielectric layer surrounding each nanosheet, and a metal gate that wraps entirely around the stacked channels [1], [2]. Modern GAA nanosheet FETs are typically fabricated using silicon (Si) or silicon-germanium (SiGe) nanosheets as the channel material due to their effective carrier transport properties. The gate dielectric is formed using high- κ materials such as hafnium dioxide (HfO_2) to achieve strong gate coupling while maintaining low leakage current. Metal gate electrodes such as titanium nitride (TiN) or tungsten (W) are commonly used to control threshold voltage through tuning but come at an expensive cost. Source and drain contact regions are usually heavily doped silicon. Ongoing research is also exploring III-V compound semiconductor nanosheets for future nodes to further enhance electron mobility and reduce

operating voltage [1], [3], [5]. The vertical stacking of these nanosheets allows the effective channel width to increase without expanding the lateral footprint [1], [3]. As a result, higher drive current can be achieved within the same layout area. This new geometry is the key advantage that distinguishes nanosheet FETs from planar MOSFETs, FinFETs, TreeFETs, and CombFETs.

A. Principles of Operation

The GAA nanosheet FET operates on the same fundamental principle as a conventional MOSFET. When a voltage is applied to the gate terminal, an electric field forms across the thin gate dielectric and enters the nanosheet channels. For an n-type device, a positive gate voltage attracts electrons toward the surface of each nanosheet forming an inversion layer that allows current to flow from the source to the drain. This is unlike planar MOSFETs and FinFETs, where the gate controls only one or three sides of the channel. As stated in II. Structure and Function, the defining advantage of the GAA architecture is that the gate completely surrounds each nanosheet channel on all four sides. This results in more ideal electrostatic behavior between the gate and the channel. This allows the potential barrier between the source and drain to be controlled exclusively by the gate voltage rather than by the drain voltage. This significantly suppresses short-channel effects such as drain-induced barrier lowering (DIBL), threshold voltage roll-off, and excessive off-state leakage current [1], [3], [8].

In addition, the complete gate enclosure leads to a steeper subthreshold slope. This means the device can more quickly switch between its off state and on state. This enables reliable operation at lower supply voltages meaning reduced power consumption. Since multiple nanosheets are stacked vertically and operate in parallel, the total drain current is the sum of the currents flowing through each individual sheet [2], [3]. This allows the device to achieve high drive current without increasing lateral device dimensions.

B. Function

The primary function of the GAA nanosheet FET is to serve as the fundamental switching device for logic and memory circuits where standard FinFETs begin to fail. As devices scale below 5 nm, limitations in electrostatic control, drive current, and power efficiency become increasingly worse for traditional transistor geometries. The vertically stacked nanosheet structure addresses these challenges by enabling both high drive strength and low leakage within a smaller horizontal footprint.

Recent research demonstrates that by adjusting nanosheet width and stacked-channel count, designers can tune the balance between performance and power consumption within a single device architecture [2], [3]. This capability is especially promising for modern applications such as high performance processors, artificial intelligence, and low power systems where different circuitry demands different performance/energy tradeoffs [6], [7]. The vertical channel stacking allows higher on state current without increasing lateral device area. This improves switching speed and logic gate drive.

Beyond replacing FinFETs, nanosheet FETs also outperform alternative multi-fin architectures such as TreeFETs and CombFETs at the circuit level [4].

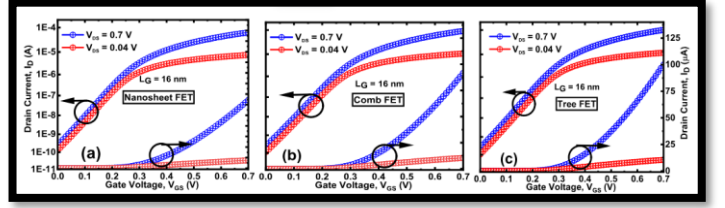


Figure 2: Simulated transfer characteristics (I_D - V_{GS}) of (a) nanosheet FET, (b) CombFET, and (c) TreeFET at $L_G = 16\text{nm}$ for two drain biases [4]. The nanosheet FET exhibits improved subthreshold behavior and reduced leakage compared to CombFET and TreeFET, indicating stronger electrostatic control and better low-power performance.

A promising future research direction for GAA nanosheet FETs is the optimization of nanosheet width, vertical stack count, and gate dielectric engineering to tune performance at the circuit level. While this work demonstrated that drive current scales nearly linearly with effective channel width and stacked-channel count, future designs could vary stack height across different circuit blocks to optimize the speed and power tradeoffs within the same chip. Additional opportunities exist in exploring alternative dielectrics to further reduce operating voltage while maintaining a higher drive current. These strategies that simultaneously optimize nanosheet geometry and interconnect properties could allow for lower energy demanding systems that new processors and memory systems can utilize. These advantages position GAA nanosheet FETs as the most promising architecture for future logic and memory technologies.

III. PERFORMANCE/SIMULATIONS

The primary performance metrics used to evaluate GAA nanosheet field-effect transistors include drive current (I_{ON}), subthreshold swing (SS), drain-induced barrier lowering (DIBL), switching speed, and energy efficiency. Drive current is especially critical because it directly determines logic gate delay and dynamic circuit performance. Since nanosheet FETs achieve increased current through both nanosheet width and stacked-channel count, current scaling behavior forms the central focus of this performance section.

To study this behavior, transfer characteristics were measured using device simulations performed on a baseline n-type MOSFET structure with fixed physical dimensions and material parameters (could not find a working nanosheet FET simulation). Although the simulator models lateral width scaling, the results directly map to vertical nanosheet stacking through effective channel width equivalence. Drive current was measured at the highest applied gate voltage for each device width and used as the baseline for stacked-channel projections.

Simulation results show that drive current increases nearly linearly with effective channel width [2], [3]. A 200 nm wide device produced $I_{ON} \approx 27.1 \mu\text{A}$, while 400 nm, 600 nm, and 800 nm devices produced approximately 54.2 μA , 81.3 μA , and 108.4 μA respectively.

This confirms the fundamental proportionality:

$$I_{ON} \propto W$$

In GAA nanosheet FETs, stacked vertical channels increase the effective width without increasing lateral footprint. Therefore, the total drive current scales as:

$$I_{ON,total} = N \cdot I_{ON,single\ sheet}$$

where N is the stacked-channel count. This property enables nanosheet FETs to achieve significantly higher drive current than FinFETs, TreeFETs, or CombFETs while maintaining better electrostatic control in a more compact footprint [1]-[4].

In nanosheet FETs, each nanosheet acts as an independent conduction channel controlled by the surrounding gate. Since these channels operate in parallel, the total drive current is the sum of the individual sheet currents. The effective channel width in a nanosheet device is therefore given by:

$$W_{eff} = N \cdot W_{sheet}$$

which leads directly to:

$$I_{ON,total} = N \cdot I_{ON,single\ sheet}$$

Although the device simulator used in this work models a conventional MOSFET rather than a vertically stacked nanosheet structure, the simulation still demonstrates that drive current increases linearly with effective channel width. This linear width/current proportionality directly translates to vertical stacked-channel scaling in nanosheet FETs.

Using the measured baseline simulation data for a 200 nm wide device with $I_{On} \approx 27.1 \mu A$, equivalent stacked-channel scaling yields:

- $N = 2 \rightarrow I_{On} \approx 54.2 \mu A$
- $N = 3 \rightarrow I_{On} \approx 81.3 \mu A$
- $N = 4 \rightarrow I_{On} \approx 108.4 \mu A$

Device Width (nm)	Extracted I_{On} (μA)
200	27.1
400	54.2
600	81.3
800	108.4

Figure 3: Simply Compiled Simulation Data

These values match the direct width scaled simulation results, confirming that stacked nanosheets provide a nearly ideal linear current scaling mechanism.

TreeFETs and CombFETs attempt to improve drive current by introducing multiple laterally branched or interdigitated conduction fins. While these geometries increase the number of conduction pathways, they do so at the cost of:

- Increased parasitic gate and fringe capacitance
- Reduced electrostatic control
- Greater layout complexity

These parasitic effects degrade switching speed and raise the energy-delay product (EDP) at the circuit level. At this circuit

level, Kumari *et al.* demonstrate that stacked GAA nanosheet FET inverters consistently achieve lower energy-delay product than both TreeFET and CombFET implementations due to reduced parasitic capacitance and better electrostatic gate control [4]. While each of which have an equal footprint layout as well.

In contrast, nanosheet FETs increase current through vertical channel stacking, not lateral layout growth. This preserves compact footprint, minimizes parasitic capacitance increase, and maintains near-ideal gate control since each sheet remains fully surrounded by the gate. As a result, nanosheet FETs achieve [4], [7]:

- Higher switching speed
- Lower dynamic energy per transition
- Reduced leakage power

All this leads to nanosheet FETs outperforming TreeFETs and CombFETs not merely at a device level, but also at the circuit level, where capacitance, delay, and energy efficiency dictate performance.

IV. THE SIMULATION METHOD

To evaluate how drive current scales with effective channel width and stacked-channel count, a series of device simulations were performed using the nanoHUB MOSFET Lab (ABaCUS) simulator (*This is the data that was previously explained in III. Performance/Simulations*). A baseline n-type MOSFET structure was used to model current transport under fixed physical and material parameters. While this simulator models lateral channel width rather than a true vertically stacked GAA nanosheet structure, the results directly represent effective width scaling, which maps one-to-one to stacked nanosheet scaling through parallel-channel equivalence.

All device parameters were held constant across all simulation runs to ensure that only effective channel width influenced the observed current scaling. The channel length was set to 20 nm, oxide thickness to 2 nm, and junction depth to 20 nm. The source and drain lengths were fixed at 50 nm, and the substrate thickness was set to 50 nm. Drift-diffusion transport was used with vertical and parallel field-dependent mobility enabled. The operating temperature was held at 300 K for all simulations.

For each simulation run, the device width was adjusted while all other parameters remained unchanged. Four separate widths were simulated: 200 nm, 400 nm, 600 nm, and 800 nm. For each width, an IdVg transfer characteristic was generated with the gate voltage swept from 0 V to 1.0 V using 20 evenly spaced bias points. The drain voltage was swept between 0.1 V and 1.0 V to satisfy simulator constraints, producing multiple transfer curves. The bulk bias was held at 0 V.

Drive current (I_{On}) was extracted at the highest applied gate voltage for each simulation and used as the reference value for current scaling analysis. The extracted single-width current values were then used to analytically model vertically stacked nanosheet operation by scaling the drive current proportionally with stacked-channel count.

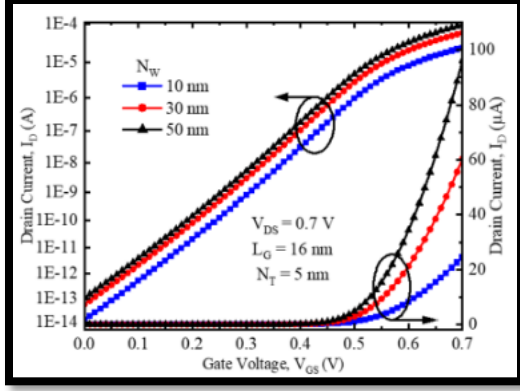


Figure 4: Published transfer characteristics (I_D - V_{GS}) of a GAA nanosheet FET for multiple nanosheet widths, showing increased drive current I_{ON} with increasing width at $V_{DS} = 0.7$ V [3], consistent with the width-scaling trend observed in our simulations.

V. SIMULATION ANALYSIS AND LIMITATIONS

The simulation results clearly demonstrate that drive current scales linearly with effective channel width. For the 200 nm wide device, the extracted drive current at maximum gate bias was approximately 27.1 μ A. When the width was doubled to 400 nm, the drive current increased to approximately 54.2 μ A. Further increases in width to 600 nm and 800 nm yielded drive currents of approximately 81.3 μ A and 108.4 μ A, respectively. These results confirm a near-ideal proportional relationship between channel width and drive current.

This linear scaling behavior verifies the fundamental relationship (*As explained in III. Performance/Simulations*):

$$I_{ON} \propto W$$

Because GAA nanosheet FETs achieve increased effective channel width through **vertical stacking rather than lateral expansion**, this same proportionality applies directly to stacked-channel devices. Each additional nanosheet acts as an independent parallel conduction path, giving:

$$I_{ON,total} = N \cdot I_{ON,single\ sheet}$$

Overall, the simulation results validate the main thesis of this project in which drive current scales linearly with effective nanosheet width and stacked-channel count. This vertical scaling mechanism is the main reason nanosheet FETs outperform TreeFETs and CombFETs in modern circuit-level performance and energy efficiency.

The device simulations performed in this work model a planar MOSFET structure rather than a real three-dimensional GAA nanosheet transistor. As a result, the simulations directly demonstrate lateral width scaling of drive current, instead of vertical nanosheet stacking. However, the fundamental physics governing both structures is identical. The drive current is proportional to the number of parallel conduction paths available in the channel.

In GAA nanosheet FETs, vertical stacking simply provides parallel conduction paths in the out-of-plane direction rather than through lateral widening. Therefore, the linear width-

current proportionality observed in the simulations directly maps to stacked-channel scaling under the effective width equivalence:

$$W_{eff} = N \cdot W_{sheet}$$

where N is the number of nanosheets. This approach is consistent with how stacked-channel scaling is treated in industrial modeling.

Futhermore, the simulations operate under drift-diffusion transport assumptions and do not explicitly capture ballistic transport, quantum confinement, or parasitic source-drain resistance effects that become important at further scaled levels. Nevertheless, the results accurately capture first-order scaling trends and are sufficient for validating the fundamental drive-current scaling behavior central to this project's "deep-dive" analysis.

VI. PROPOSED NEW DIRECTION

Based on these results, using vertical stack scaling could produce a revolutionary way of computing low voltage logic.

Stack Count	Projected I_{ON} (μ A)
1	27.1
2	54.2
3	81.3
4	108.4
5	135.5
6	162.6
7	189.7

Figure 5: Potential New Direction Data (This data was initially recorded from a MOSFET simulation, not a nanosheet FET simulation. Although the physics works the same and the trend will compare to this data, an actual nanosheet FET simulation will provide different values).

These projections indicate that by increasing the vertical stack count from four to six nanosheets, drive current can be increased by approximately 50% without any increase in lateral device footprint. This added drive strength could be directly traded for reduced supply voltage while preserving switching speed, enabling low voltage/energy logic computing for the next generation of computers.

A huge challenge for this continued stacking method is increased source-drain series resistance, thermal dissipation, and fabrication complexity. Future work could address these limitations through advanced contact engineering and optimized GAA methods. Regardless, these analytical results indicate that extended vertical stacking represents a promising research direction for improving the energy-delay product of GAA nanosheet FETs beyond current technology limits.

VII. CONCLUSION

This report examined the structure, operation, and performance of Gate-All-Around (GAA) nanosheet field-effect transistors and explained why this architecture represents the most promising path forward for transistor scaling beyond the limits of current FinFET technology. By fully surrounding each conduction channel with the gate electrode and enabling vertical stacking of nanosheets, GAA devices achieve superior

electrostatic control, suppression of short-channel effects, and scalable channel width without increasing lateral footprint.

A deep dive into drive current scaling demonstrated that current increases nearly linearly with effective channel width and stacked-channel count. Device simulations performed using the nanoHUB MOSFET Lab simulator confirmed this behavior by explicitly showing linear scaling of drive current with lateral device width. Through effective width equivalence, these results directly translate to vertical nanosheet stacking, where each additional nanosheet provides an independent parallel conduction path. The recorded simulation results showed that a four-channel equivalent device achieves nearly four times the drive current of a single channel device.

The analysis also illustrated why nanosheet FETs outperform alternative multi-fin architectures such as TreeFETs and CombFETs at the circuit level. While those devices increase current through lateral branching, they suffer from increased parasitic capacitance and reduced gate efficiency. In contrast, vertical nanosheet stacking preserves the compact footprint, minimizes parasitics, and maintains better gate control, leading to a lower energy-delay product and faster switching speed.

Overall, the results of this study demonstrate that GAA nanosheet FETs provide a better, although complex, balance of performance, power efficiency, and scalability compared to current traditional transistor architectures. With continued advances in fabrication and materials, nanosheet FET technology is positioned to remain the dominant logic device architecture for the future generations of integrated circuits.

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